ECE HPC Cluster

Introduction

ECE HPC Cluster provides primarily GPU computing services similar to ITSC HPC research computing. The main difference is that ECE HPC cluster virtual based which is built on virtualization technologies (VMware vSphere hypervisor, Software Defined Storage (VSAN and CEPH).

Here are the main characteristics of ECE HPC cluster:

- Aimed to maximize the GPU resource utilization. To achieve the goal, we use Slurm (Simple Linux Utility for Resource Management) to distribute user jobs to backend virtual GPU nodes for execution.
- Principle Investor (PI) will have a dedicated Slurm resource partition (queue) with virtually no limit in job execution time.
- PI can at least use the GPU resource he/she has invested. In addition, PI can use idle GPU resources in the cluster.
- If PI owned resource is in use by other groups, that occupying program (lower priority) will be preempted by the PI group (higher priority), and resume running when the resource becomes available.

Quick User Guide

Access the ECE HPC Cluster

You can access the **hpc.ece.ust.hk**, the login-node of the cluster with Secure Shell (SSH). On PC Windows platform, you can use the free SSH client like PuTTY or MobaXterm.

```
ssh  user@hpc.ece.ust.hk
```

Please ensure that your ITSC account is enrolled with two-factor authentication.

You can refer to the following link for more details. Please change login node to hpc.ece.ust.hk.
HKUST SuperPOD - How to Login | Information Technology Services Center

Batch job submission

Recommended way for running GPU computationally intensive, non-interactive jobs, e.g. training AI models.

1. Connect login node **hpc.ece.ust.hk** with ITSC account using SSH client
2. Create a job script for submission. Here is a sample sbatch job script named "stest_tf.sbatch" to request 1 GPU to run a python script.

stest_tf.sbatch:

```
#!/bin/bash

# Job name
#SBATCH  --job-name=stest_tf
# Job standard output file
#SBATCH  --output=stest_tf.%j

# Email notification type: job start and finish
#SBATCH  --mail-type=all
# Update with your ITSC email address!!!
#SBATCH  --mail-user=username@ust.hk

# Request a specific partition for the resource allocation
# Consult computing team for your_PI_group
#SBATCH  -p gpu-share,your_PI_group

# Request 20 mins execution time (format: "day-hour:min:sec")
#SBATCH  --time=00:20:00
# Request 1 GPU
#SBATCH  --gres=gpu:1

# Setup runtime environment if necessary
module  purge
module  load anaconda3/2022.10

# Run the python code
python  ./stest_tf.py
```

stest_tf.py:

```
import tensorflow as tf

# Check if GPU is available
physical_devices = tf.config.list_physical_devices('GPU')
if len(physical_devices) == 0:
    print("No GPU available.")
else:
    tf.config.experimental.set_memory_growth(physical_devices[0], True)

# Create a tensor on the GPU
with tf.device('/GPU:0'):
    tensor = tf.random.normal((10000, 10000))

    # Perform some heavy computation on the GPU
    result = tf.matmul(tensor, tensor)

# Print the result
print(result)
```

3. Submit to the cluster with command:

   sbatch stest_tf.sbatch

Here are some commonly used commands for inquiring and managing your jobs.

Get information about the cluster:	**sinfo**
Check your running job status:	**squeue --me**
Cancel your job:	**scancel job_number**

Interactive Job mode

Apart from submitting batch job to the cluster, user can acquire CPU / GPU resource from cluster interactively. The best use case is to test run a program or script with a small amount of data for the purpose of proof of concept and/or program debugging. Once test is successful the program will be submitted in batch mode (sbatch) with full scaled data.

Example 1. Request a GPU node to run a python script stest_tf.py.

```
# Request partition "gpu-dev" (MIG 1g.10gb)
user@ecenod198:~$ srun  --pty  -p gpu-dev  --gres gpu  /bin/bash

# Once requested resource is obtained, you will be assigned a compute node
(ecenod233).

# Prepare the environment
user@ecenod233:~$ module  purge
user@ecenod233:~$ module  load  anaconda3/2022.10

# Run the python code
user@ecenod233:~$ python  ./stest_tf.py

# Release the resource after use. Leave the compute node and return to login
node.
user@ecenod233:~$ exit
user@ecenod198:~$
```

Example 2. Request a GPU node to run interactive program jupyter-lab.

1. Setup jupyter-lab server in the cluster

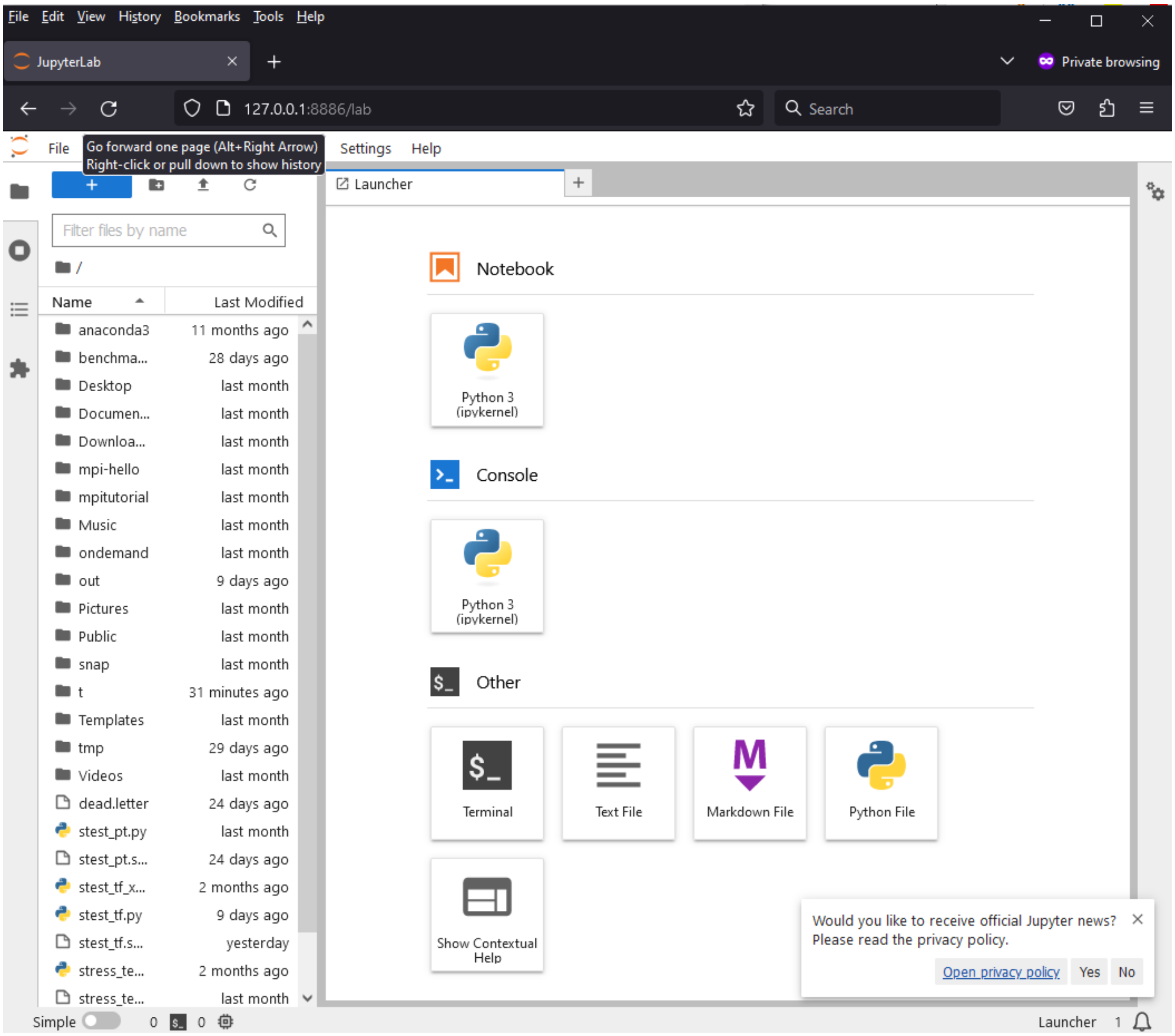
```
# Request partition "gpu-dev" (MIG 1g.10gb)
user@ecenod198:~$  srun  --pty  -p gpu-dev  --gres gpu  /bin/bash

# Once requested resource is obtained, you will be assigned a compute node
e.g: (ecenod233).

# Prepare the environment
user@ecenod233:~$ module  purge
user@ecenod233:~$ module  load  anaconda3/2022.10

# Start jupyter lab server. Choose a free port (8886)
user@ecenod233:~$ jupyter-lab --no-browser --ip 0.0.0.0 --port 8886

# A connection URL will be generated and shown near end of the dialog, e.g.:
http://127.0.0.1:8886/lab?token=9639cf533fab8a29633e1dd0fb1106778da97746d34
964dx
```

2. In your desktop computer, build a ssh forwarding rule:

```
ssh  -L  8886:ecenod208:8886  user@hpc.ece.ust.hk
```

3. Open the generated URL above with a web browser:
   http://127.0.0.1:8886/lab?token=9639cf533fab8a29633e1dd0fb1106778da97746d34964dx

The jupyter-lab UI is shown:



4. Remember to release the resource in the cluster after use.

```
# Ctrl-C to stop Jupyter-lab server in assigned compute node.
Shutdown this Jupyter server (y/[n])? y

# Release the resource after use. Leave the compute node.
user@ecenod233:~$ exit
user@ecenod198:~$
```

Usage Recommendations and Guidelines

- Login node is for job submission and debug only. Never run full scale simulation job on it.
- Test run program on interactive mode (srun) with mininal dataset as a proof of concept. Once succeeded submit the job to the cluster with full scaled data.
- Choose GPU partition(gpu-share, gpu-dev, your_PI_group) for GPU workloads, and CPU partition(cpu-share) for CPU workloads for best resource utilization.
- Due to similar technology with ITSC HPC Cluster, you can refer to the following links.
  HPC3 Cluster User Guide & FAQ | Information Technology Services Center (hkust.edu.hk)
  HKUST SuperPOD - Use of Slurm Job Scheduling System | Information Technology Services Center